Xiaopu Wang

State College, PA • xmw5221@psu.edu
shooof.github.io/homepage • github.com/shooof

Education

Pennsylvania State University — 2024 – Present
Ph.D. in Statistics; Advisor: Runze Li — *University Park, PA*

University of Science and Technology of China (USTC) — 2021 – 2024
M.S. in Statistics — *Hefei, China*

Nankai University — 2017 – 2021
B.S. in Mathematics — *Tianjin, China*

Publications

- **Wang, X.**, He, Z., Liu, C., and Li, R. (2026). *Beyond Heuristic Tuning: Power-Calibrated LLM Watermarking*. **International Conference on Machine Learning (ICML)**.
- *Jiang, Y., *Zhao, B., ***Wang, X.**, *Tang, B., *et al.* (*co-first) (2025). *UKB-MDRMF: A Multi-Disease Risk and Multimorbidity Framework Based on UK Biobank Data*. **Nature Communications**. doi:10.1038/s41467-025-58724-3.
- **Wang, X.**, Wang, X., Zhang, A., and Wen, C. (2024). *Copy Number Variation Detection Based on Constraint Least Squares*. **Statistics and Its Interface**, 17(1), 27–37. doi:10.4310/23-SII814.
- Shen, Y., Wang, J., . . . , **Wang, X.**, *et al.* (2025). *CATI: A Medical Context-Enhanced Framework for Diagnosis Code Assignment in the UK Biobank Study*. **Artificial Intelligence in Medicine**. doi:10.1016/j.artmed.2025.103136.

Research Experience

Power-Calibrated LLM Watermarking — 2024 – Present
Pennsylvania State University — *University Park, PA*

- Derived closed-form relationships connecting logit-based watermark parameters to green-token rates, detection power, and KL distortion.
- Converted parameter selection into a constrained optimization problem and demonstrated stronger power–distortion trade-offs than heuristic baselines across four model families and three text domains.

Marked Hawkes Processes for Recurrent Events in EHRs — 2025 – Present
Pennsylvania State University; collaboration with UNC — *University Park, PA*

- Developing marked temporal point-process models that integrate longitudinal EHR events to estimate recurrent disease risks and patient trajectories.
- Built cohort-construction, preprocessing, and evaluation pipelines for real-world analysis of MIMIC-IV data.

Multi-Disease Risk and Multimorbidity Modeling — 2021 – 2024
University of Science and Technology of China — *Hefei, China*

- Built large-scale data cleaning, feature-engineering, disease-prediction, and survival-analysis pipelines for UK Biobank.
- Developed interpretable analyses and visualizations of disease risk factors, multimorbidity patterns, and temporal disease progression.

Copy Number Variation Detection — 2021 – 2024
University of Science and Technology of China — *Hefei, China*

- Formulated copy number variation detection as a constrained least-squares problem and designed an alternating minimization algorithm.
- Validated the method on simulated and real genomic data, demonstrating robust change-point recovery under heavy-tailed noise.

Technical Skills

Programming: Python, R, C++, SQL
Tools: Linux/shell, Git, LaTeX, Mathematica
Machine Learning: PyTorch, scikit-learn, XGBoost, LightGBM, CNNs, RNNs/LSTMs, Transformers, diffusion models
Statistics & Data: Hypothesis testing, survival analysis, recurrent-event modeling, causal inference, large-scale data processing, visualization
LLMs & NLP: LLM watermarking, prompt engineering, BERT, GPT-family models, retrieval-augmented generation
Biomedical Data: UK Biobank, MIMIC-IV, All of Us, clinical NLP, comorbidity analysis